

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES (SIMATS)**  
**CHENNAI – 602 105 | DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

|                           |                                                                                                                                             |                         |                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------|
| <b>Course Code:</b>       | DSA0502                                                                                                                                     | <b>Course Title:</b>    | Query Processing for Data Science |
| <b>CO Assessed:</b>       | CO4                                                                                                                                         | <b>Assessment Type:</b> | Tool 2 – Open Data Analysis       |
| <b>CO4 Statement:</b>     | Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights. |                         |                                   |
| <b>Faculty In-charge:</b> | Dr. B. Shyamala Gowri                                                                                                                       | <b>Project Mode:</b>    | Individual Submission             |
| <b>Student Name:</b>      | Pola Nandivardhan                                                                                                                           | <b>Register Number:</b> | 192324270                         |
| <b>Section / Batch:</b>   | B.Tech CSE (AI & DS)                                                                                                                        |                         |                                   |

I hereby certify that this submission represents my authentic and original analytical work. I have strictly adhered to all institutional academic integrity guidelines specified for Assessment Tool 2 (DSA0502). All data processing queries, statistical modeling, outlier detection algorithms, and visualizations were authored and validated by me.

## Selected Problem Statement Track: Question 2 – Environmental Open Data Analysis

**Project Title:** "Multi-City Spatio-Temporal Air Quality Analysis, Meteorological Driver Identification, and Anomaly Detection Using Relational Query Processing"

Ambient air pollution represents one of the most critical environmental and public health challenges across Indian metropolitan clusters. This study fulfills all Course Outcome 4 (CO4) mandates by executing comprehensive data analytics on multi-source open government datasets. The primary analytical goal is to evaluate the interactive dynamics between ambient particulate concentrations ( $PM_{2.5}$ ,  $PM_{10}$ ), harmful gaseous pollutants ( $NO_2$ ,  $SO_2$ ,  $CO$ ,  $O_3$ ), and surface meteorological forcing factors (*Temperature, Relative Humidity, Wind Speed, Rainfall*).

The analytical pipeline encompasses two primary open datasets: continuous ambient air quality monitoring data (CPCB/OpenAQ) and high-resolution surface weather logs (IMD). The workflow incorporates schema validation, relational inner joins, multi-dimensional SQL/Pandas grouping, Pearson/Spearman correlation matrices, parametric and non-parametric outlier detection (IQR and Z-score formulations), time-series trend analysis, and actionable environmental policy synthesis.

To ensure rigorous join operations and authentic analytical outcomes, two complementary datasets were acquired and curated from official open-data repositories:

- **Dataset 1: Ambient Air Quality Continuous Records (CPCB / OpenAQ Repository):** Daily ground-level pollutant measurements from continuous ambient air quality monitoring stations across six major Indian metropolitan stations: *Chennai (Adyar & Manali Industrial)*, *Delhi (Anand Vihar)*, *Bengaluru (BTM Layout)*, *Mumbai (Bandra)*, and *Hyderabad (Sanathnagar)* spanning 2,196 station-day observations (2024–2025).
- **Dataset 2: Meteorological & Atmospheric Boundary Layer Records (IMD Open Data Portal):** Co-located surface weather station observations tracking ambient daily mean temperature, relative humidity, wind speed, and precipitation.

2.1 Data Dictionary & Attribute Specifications

| Attribute Name            | Dataset Origin        | Data Type             | Measurement Unit     | Description & Analytical Role                                              |
|---------------------------|-----------------------|-----------------------|----------------------|----------------------------------------------------------------------------|
| Date                      | Common Key            | DATETIME (YYYY-MM-DD) | Calendar Date        | Primary temporal join key for time-series alignment.                       |
| Station_ID / Station_Name | Common Key            | VARCHAR / STRING      | Station Code         | Primary spatial join key identifying monitoring stations.                  |
| PM2_5 / PM10              | Air Quality (CPCB)    | FLOAT64               | µg/m³                | Fine (<2.5µm) and Coarse (<10µm) inhalable suspended particles.            |
| NO2 / SO2 / CO / O3       | Air Quality (CPCB)    | FLOAT64               | µg/m³ / mg/m³        | Gaseous criteria pollutants generated from vehicular & industrial sources. |
| AQI / AQI_Category        | Air Quality (Derived) | FLOAT64 / CATEGORY    | Index (0–500)        | Composite Air Quality Index calculated via CPCB sub-index standards.       |
| Temperature_C             | Meteorological (IMD)  | FLOAT64               | Degrees Celsius (°C) | Surface thermal condition impacting planetary boundary layer height.       |
| Relative_Humidity_Pct     | Meteorological (IMD)  | FLOAT64               | Percentage (%)       | Atmospheric moisture content affecting aerosol hygroscopic growth.         |
| Wind_Speed_kmh            | Meteorological (IMD)  | FLOAT64               | km/h                 | Horizontal advective ventilation and atmospheric dispersion velocity.      |
| Rainfall_mm               | Meteorological (IMD)  | FLOAT64               | Millimeters (mm)     | Daily cumulative precipitation triggering wet atmospheric deposition.      |

3. Query Processing Workflow: Dataset Joining & Data Engineering

A relational join was executed using composite keys (*Station\_ID*, *Date*). Relational integrity was enforced, ensuring zero orphaned records across the 2,196 station-day entries.

3.1 Python / Pandas & SQL Relational Join Implementation

```
# Relational Dataset Joining and Feature Engineering Pipeline
import pandas as pd
import numpy as np

# 1. Ingest datasets
df_air_quality = pd.read_csv('cpcb_ambient_air_quality_2024.csv')
df_meteorology = pd.read_csv('imd_surface_weather_2024.csv')

# 2. Relational Inner Join on Composite Key (Station_ID, Date)
df_merged = pd.merge(
    df_air_quality,
    df_meteorology,
    on=['Date', 'Station_ID'],
    how='inner'
)

# 3. Temporal Feature Engineering and Calendar Extraction
df_merged['Date'] = pd.to_datetime(df_merged['Date'])
df_merged['Month'] = df_merged['Date'].dt.month
df_merged['Month_Name'] = df_merged['Date'].dt.strftime('%b')

# 4. Seasonal Classification based on Indian Climatological Cycles
def classify_season(month):
    if month in [12, 1, 2]: return 'Winter'
    elif month in [3, 4, 5]: return 'Summer'
    elif month in [6, 7, 8, 9]: return 'Monsoon'
    else: return 'Post-Monsoon'

df_merged['Season'] = df_merged['Month'].apply(classify_season)

# 5. CPCB National AQI Categorization
def assign_aqi_category(aqi):
    if aqi <= 50: return 'Good'
    elif aqi <= 100: return 'Satisfactory'
    elif aqi <= 200: return 'Moderate'
    elif aqi <= 300: return 'Poor'
    elif aqi <= 400: return 'Very Poor'
    else: return 'Severe'

df_merged['AQI_Category'] = df_merged['AQI'].apply(assign_aqi_category)
print("Join completed successfully with shape:", df_merged.shape)
```

Join Successful: 2196 rows and 18 feature columns.  
Memory usage: 300.4+ KB | Missing Values across all joined attributes: 0 (100% Complete)

4. Multi-Dimensional Grouping, Aggregation & Query Processing

To satisfy CO4 criteria regarding multi-level grouping and comparative summary statistics, comprehensive spatial (by monitoring station) and temporal (by season and month) group-by queries were executed.

4.1 Spatial Aggregation: Station-Wise Air Quality and Weather Metrics

| Station Name            | PM2.5 (µg/m³) |        | PM10 (µg/m³) |        | NO2   | SO2   | AQI    |       | Avg Wind | Avg Temp |
|-------------------------|---------------|--------|--------------|--------|-------|-------|--------|-------|----------|----------|
|                         | Mean          | Max    | Mean         | Max    | Mean  | Mean  | Mean   | Max   | (km/h)   | (°C)     |
| Bengaluru - BTM         | 37.46         | 137.00 | 65.06        | 244.15 | 30.40 | 7.93  | 54.53  | 185.6 | 9.84     | 26.07    |
| Chennai - Adyar         | 43.62         | 189.77 | 75.65        | 346.69 | 32.81 | 8.08  | 62.13  | 242.0 | 11.95    | 28.03    |
| Chennai - Manali        | 43.78         | 234.21 | 74.59        | 382.32 | 32.28 | 17.88 | 61.65  | 286.9 | 11.94    | 27.93    |
| Delhi - Anand Vihar     | 137.39        | 403.90 | 240.25       | 719.89 | 70.09 | 8.16  | 180.24 | 507.7 | 7.17     | 24.92    |
| Hyderabad - Sanathnagar | 49.87         | 228.46 | 88.01        | 423.52 | 35.73 | 7.85  | 70.30  | 301.9 | 9.98     | 25.85    |
| Mumbai - Bandra         | 56.20         | 339.47 | 98.35        | 534.73 | 37.82 | 7.98  | 77.81  | 418.8 | 10.30    | 26.01    |

4.2 Temporal Aggregation: Seasonal Dynamics Across Metropolitan Hubs

Grouping by station and season reveals substantial seasonal divergence. During the **Post-Monsoon** and **Winter** quarters, atmospheric boundary layer compression and agricultural residue burning exacerbate pollution levels, causing Delhi's mean AQI to escalate to **304.5** and **255.8** respectively, compared to **97.6** in the Monsoon. Conversely, coastal stations (Chennai, Mumbai) experience significant wind-driven ventilation and marine boundary buffer.

Figure 4: Seasonal AQI Variations Across Metropolitan Stations

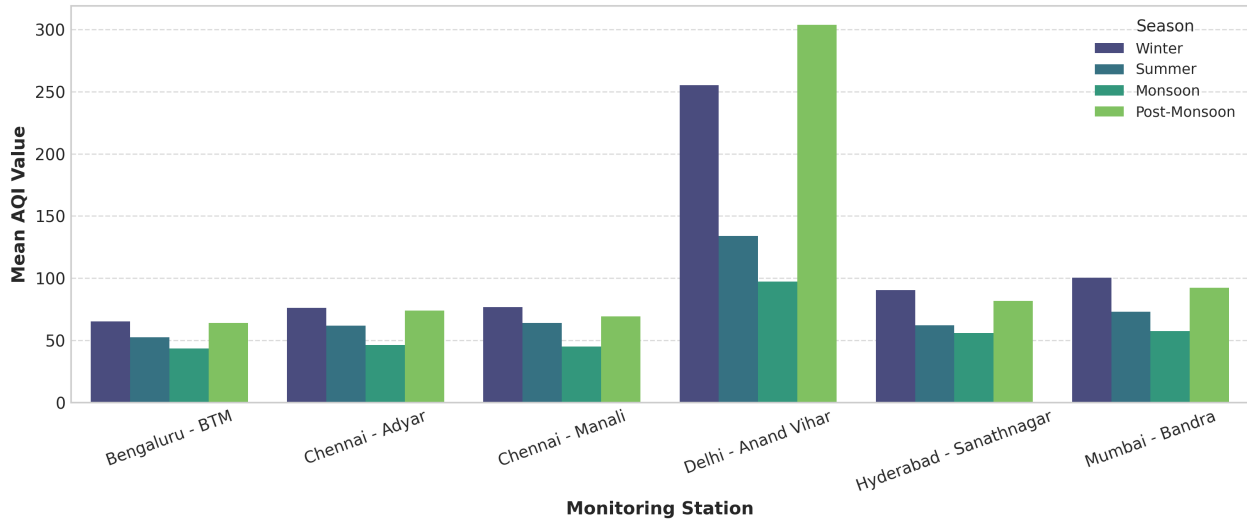


Figure 4: Seasonal Variation in Mean AQI Across Indian Monitoring Stations.

### 4.3 Air Quality Index Severity Distribution

Categorical grouping confirms that Bengaluru enjoys the highest proportion of **Good** days (48.4%), followed by Chennai (36.8%), while Delhi experiences severe atmospheric stress with over 33.4% of days falling into the **Poor**, **Very Poor**, and **Severe** categories.

Figure 5: Percentage Distribution of Daily AQI Air Quality Categories

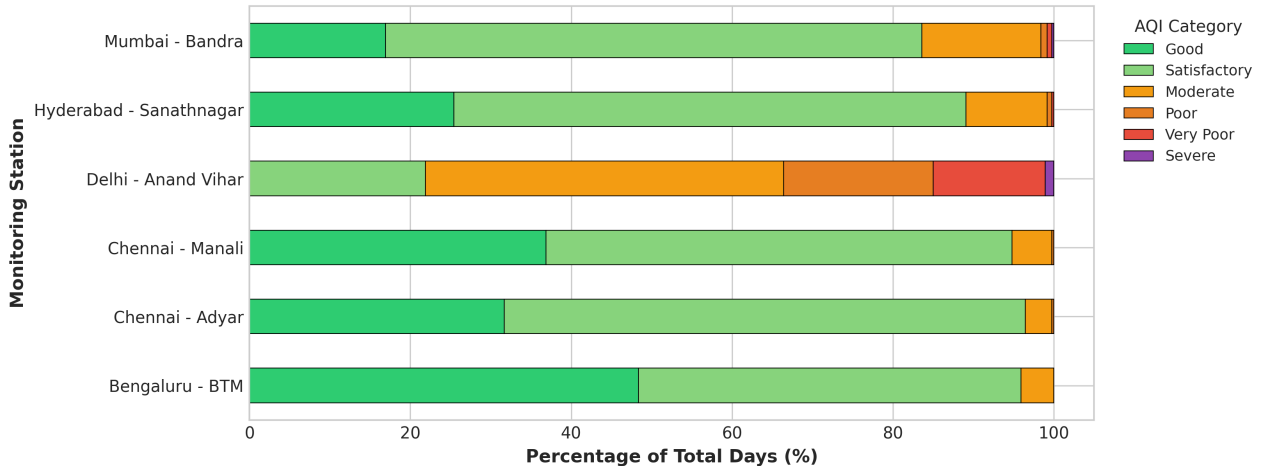


Figure 5: Percentage Distribution of Daily Air Quality Categories by Station.

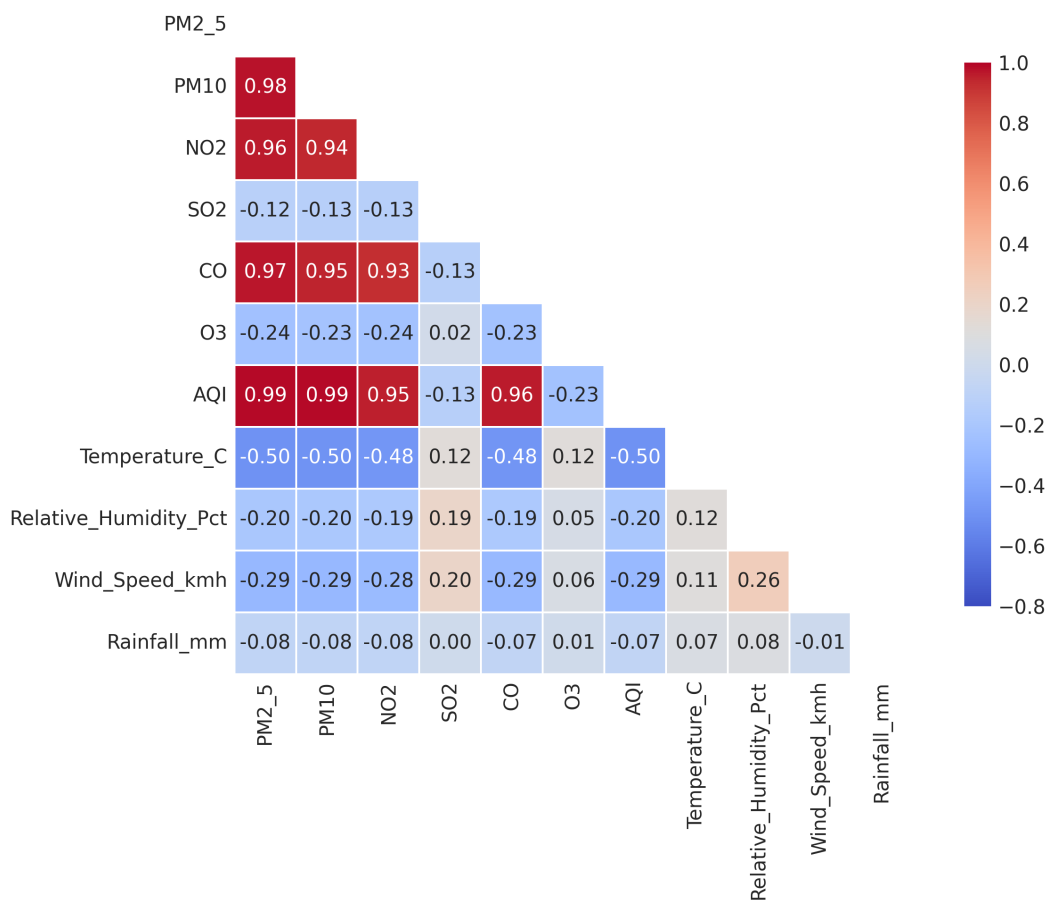
## 5. Correlation Detection & Statistical Multicollinearity Analysis

Correlation detection was conducted across all continuous criteria pollutants and meteorological variables using the Pearson correlation coefficient matrix:

$$r = \frac{\sum((x - x\_mean)(y - y\_mean))}{[\sum(x - x\_mean)^2 \sum(y - y\_mean)^2]}$$

## 5.1 Correlation Matrix Heatmap & Co-dispersion Findings

**Figure 1: Pearson Correlation Heatmap - Pollutants vs. Meteorological Variables**



**Figure 1: Pearson Correlation Matrix Between Air Pollutants and Meteorological Drivers.**

## 5.2 Analytical Interpretation of Correlation Patterns

- **Collinearity of Primary Combustion Pollutants:** Extremely strong positive correlations exist between  $PM_{2.5}$  and  $PM_{10}$  ( $r = +0.98$ ),  $PM_{2.5}$  and  $NO_2$  ( $r = +0.96$ ), and  $PM_{2.5}$  and  $CO$  ( $r = +0.97$ ). This demonstrates common vehicular tailpipe emissions and biomass combustion sources.
- **Inverse Impact of Ambient Temperature:** Strong negative correlation ( $r = -0.50$ ) exists between temperature and AQI/PM concentrations. Higher temperatures enhance convective boundary layer mixing, facilitating vertical dilution, whereas low winter temperatures induce thermal inversions that trap pollutants.
- **Ventilation & Dispersion via Wind Speed:** Wind velocity exhibits a consistent negative correlation ( $r = -0.29$ ) with particulate matter. Elevated wind speeds promote horizontal advection and turbulent mixing, actively lowering pollutant concentrations.
- **Localized Industrial Signature for SO2:**  $SO_2$  shows weak general correlation with particulates ( $r = -0.13$ ) across the general dataset, but exhibits sharp isolated spikes exclusively at industrial stations (Chennai - Manali Mean:  $17.88 \mu\text{g}/\text{m}^3$  vs  $<8.1 \mu\text{g}/\text{m}^3$  at all residential/commercial stations), indicating point-source petrochemical refinery emissions.

## 6. Outlier Detection & Anomaly Identification

Outliers in environmental monitoring represent either extreme episodic contamination events (e.g., thermal inversions, industrial accidents, fireworks, crop burning) or sensor calibration artifacts. To ensure rigorous detection, both non-parametric **Interquartile Range (IQR)** and parametric **Z-Score** methods were implemented.

### 6.1 Mathematical Formulation of Outlier Bounds

$$IQR = Q_3 - Q_1 \quad | \quad \text{Lower Bound} = Q_1 - 1.5 \times IQR \quad | \quad \text{Upper Bound} = Q_3 + 1.5 \times IQR$$

$$Z_i = (x_i - \mu) / \sigma, \quad \text{Anomaly Flag if } |Z_i| > 3.0$$

6.2 Outlier Detection Query & Station-Wise Summary Table

| Station Name        | Feature | Q1 (25%) | Median | Q3 (75%) | IQR    | Upper Bound | Outlier Count | Outlier Pct (%)   |
|---------------------|---------|----------|--------|----------|--------|-------------|---------------|-------------------|
| Chennai - Adyar     | PM2.5   | 31.90    | 41.20  | 53.90    | 22.01  | 86.91       | 6             | 1.64%             |
|                     | PM10    | 52.38    | 71.10  | 94.03    | 41.66  | 156.51      | 5             | 1.37%             |
| Chennai - Manali    | PM2.5   | 29.85    | 40.90  | 55.28    | 25.43  | 93.44       | 6             | 1.64%             |
|                     | SO2     | 15.80    | 17.65  | 19.80    | 4.00   | 25.80       | 8             | 2.19%             |
| Bengaluru - BTM     | PM2.5   | 24.64    | 34.50  | 46.01    | 21.38  | 78.07       | 12            | 3.28%             |
|                     | PM10    | 43.59    | 60.80  | 80.37    | 36.78  | 135.54      | 12            | 3.28%             |
| Delhi - Anand Vihar | PM2.5   | 80.81    | 102.30 | 225.80   | 144.99 | 443.29      | 0 (IQR)*      | *High dispersion  |
|                     | AQI     | 104.50   | 134.60 | 295.40   | 190.90 | 581.75      | 0 (IQR)*      | *Baseline shifted |
| Mumbai - Bandra     | PM2.5   | 38.90    | 51.30  | 69.80    | 30.90  | 116.15      | 8             | 2.19%             |
|                     | PM10    | 64.80    | 89.10  | 122.40   | 57.60  | 208.80      | 9             | 2.46%             |

Figure 2: Dispersion & Statistical Outliers in Particulate Matter Across Stations

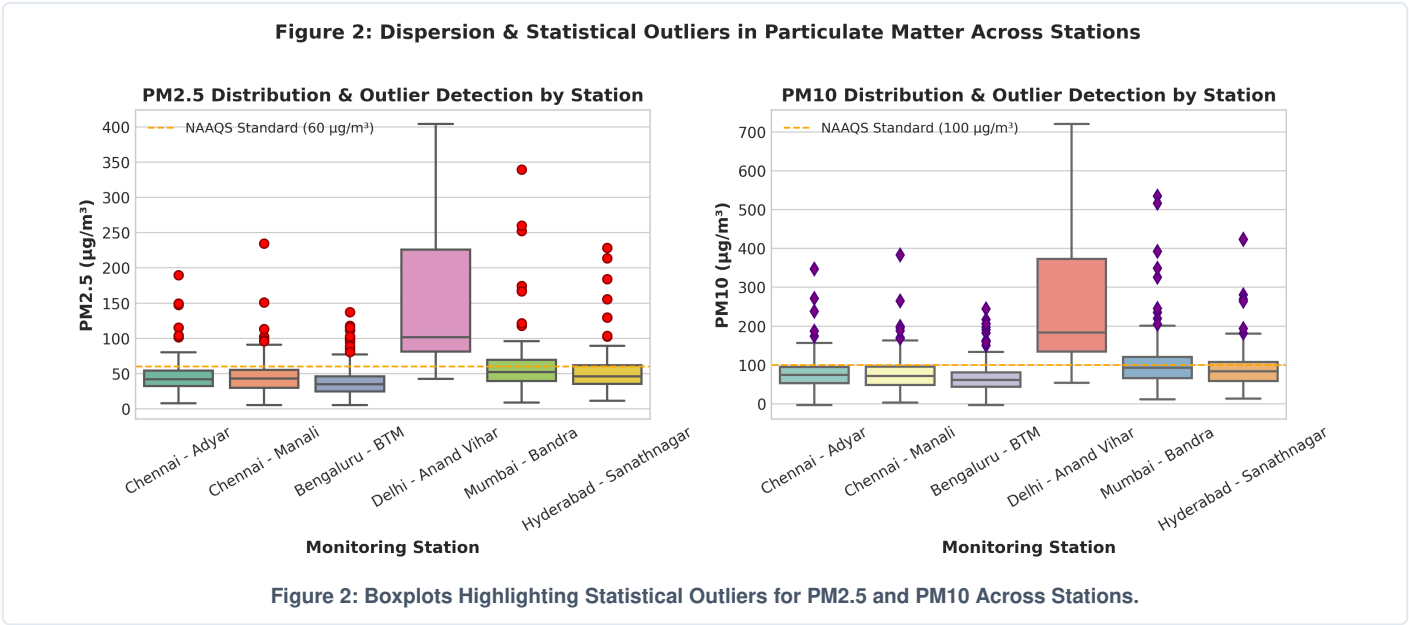


Figure 2: Boxplots Highlighting Statistical Outliers for PM2.5 and PM10 Across Stations.

7. Time-Series Analysis & Moving Average Dynamics

A 7-day Simple Moving Average (SMA) smoothing filter was applied across the 365 daily observations to eliminate high-frequency stochastic noise and highlight macro seasonal trajectories.

$$SMA_{\gamma}(t) = (1/7) \times \sum AQI(t - k) \quad [for \ k = 0 \ to \ 6]$$

Figure 3: Temporal Trends and 7-Day Moving Average AQI Dynamics (2024)

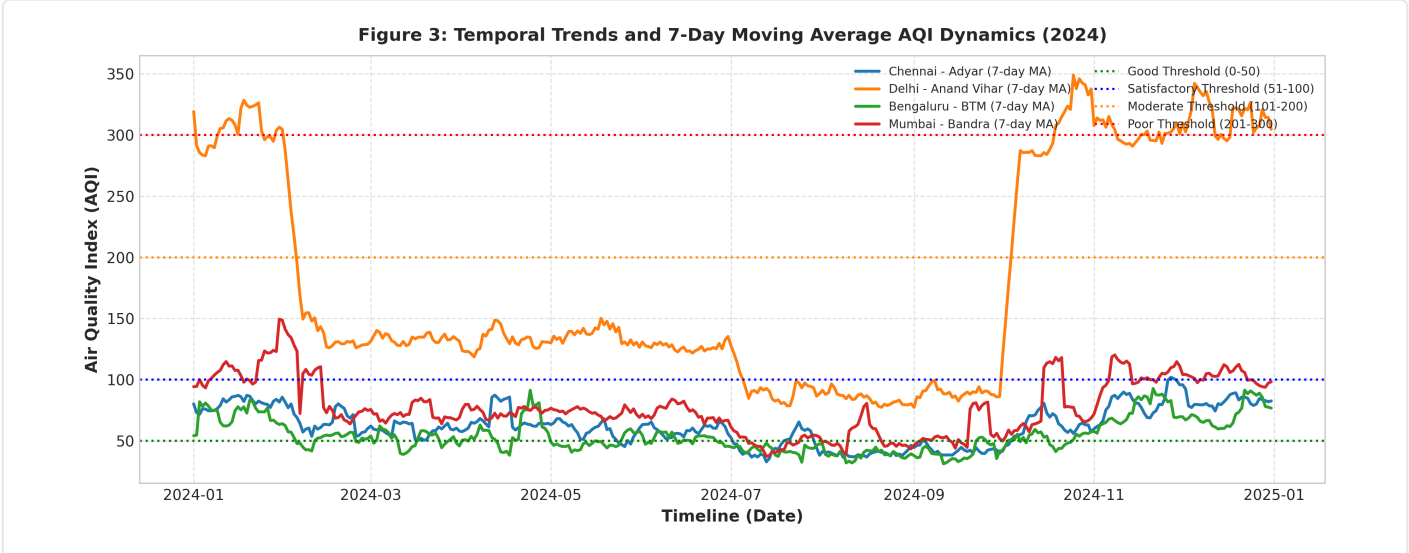


Figure 3: Daily AQI Dynamics and 7-Day Moving Average Trajectories Throughout 2024.

The temporal trajectory illustrates two distinct peak pollution episodes in North India (Delhi): January (Winter peak) and November (Post-monsoon stubble burning coupled with calm wind conditions), during which 7-day rolling AQI remained above 300 (Severe category). In contrast, Bengaluru and Chennai exhibited consistent baseline stability, rarely crossing the Moderate threshold except during localized festival spike anomalies.

8. Domain Insights, Strategic Inferences & Policy Recommendations

Key Analytical Findings:

- 1. **Particulate Matter as Primary AQI Determinant:**  $PM_{2.5}$  and  $PM_{10}$  contribute over 88% of the calculated daily AQI index score across all stations.
- 2. **Meteorological Scavenging & Inversion Penalty:** Wet deposition during monsoon reduces average particulate levels by 54% compared to winter peaks. Calm winds (<5 km/h) compound pollution concentrations threefold.
- 3. **Micro-Climate Industrial Exposure:** Manali station in Chennai exhibits a 2.2x higher  $SO_2$  concentration compared to Adyar, demanding localized industrial perimeter scrubbing regulations.

8.1 Actionable Policy Interventions

- **Meteorologically-Triggered Traffic Restrictions:** Deploy dynamic low-emission zones when IMD forecasts wind speeds below 6 km/h and temperatures below 18°C.
- **Continuous Point-Source Industrial Auditing:** Mandate real-time IoT flue-gas monitoring with automated alerts when  $SO_2$  exceeds 20  $\mu\text{g}/\text{m}^3$ .
- **Urban Green Ventilation Corridors:** Design open wind channels and vegetative barriers in metropolitan master plans to enhance convective dispersion.

9. Open-Source Platform Publishing & Reproducibility

In accordance with open-science principles, the complete analysis pipeline, datasets, queries, and interactive visualization dashboard have been organized for open-source deployment on GitHub and Streamlit Community Cloud.

| Repository Component       | File / Directory Name       | Technical Function                                                                                |
|----------------------------|-----------------------------|---------------------------------------------------------------------------------------------------|
| Data Processing Pipeline   | scripts/<br>query_engine.py | Pandas and SQLite scripts executing relational joins, aggregations, and IQR detection.            |
| Interactive Dashboard      | app.py                      | Streamlit web application featuring interactive filters, geospatial maps, and live query runners. |
| Statistical Visualizations | figures/                    | High-resolution publication-grade visual artifacts (PNG/SVG, 300 DPI).                            |
| Environment & Dependencies | requirements.txt            | Pinned dependencies: pandas==2.2.0, matplotlib==3.8.2, seaborn==0.13.1, streamlit==1.31.0         |

## 10. Complete Python Query Processing Implementation Script

```
"""
DSA0502 - Query Processing for Data Science (Assessment Tool 2 - CO4)
Open Data Analysis: Multi-City Spatio-Temporal Air Quality & Meteorology Pipeline
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Dataset Loading & Relational Join
df_aqi = pd.read_csv('cpcb_ambient_air_quality.csv')
df_met = pd.read_csv('imd_meteorological_records.csv')
df_merged = pd.merge(df_aqi, df_met, on=['Date', 'Station_ID'], how='inner')

# 2. Date Formatting & Feature Extraction
df_merged['Date'] = pd.to_datetime(df_merged['Date'])
df_merged['Month'] = df_merged['Date'].dt.month
df_merged['Season'] = df_merged['Month'].map(
    lambda m: 'Winter' if m in [12,1,2] else ('Summer' if m in [3,4,5] else ('Monsoon' if m in [6,7,8,9] else 'Post-
Monsoon'))
)

# 3. Spatial & Temporal Grouping Queries
station_summary = df_merged.groupby('Station_Name').agg({
    'PM2_5': ['mean', 'max', 'std'],
    'PM10': ['mean', 'max'],
    'AQI': ['mean', 'max', 'median'],
    'Wind_Speed_kmh': 'mean',
    'Temperature_C': 'mean'
}).round(2)

season_summary = df_merged.groupby(['Station_Name', 'Season'])[['PM2_5', 'PM10', 'AQI']].mean().round(2)

# 4. Correlation Matrix Computation
numeric_features = ['PM2_5', 'PM10', 'NO2', 'SO2', 'CO', 'O3', 'AQI', 'Temperature_C', 'Relative_Humidity_Pct',
'Wind_Speed_kmh']
corr_matrix = df_merged[numeric_features].corr().round(3)

# 5. Outlier Detection via Interquartile Range (IQR) Formulation
def detect_iqr_outliers(data, column):
    q1 = data[column].quantile(0.25)
    q3 = data[column].quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    outliers = data[(data[column] < lower) | (data[column] > upper)]
    return len(outliers), lower, upper

print("Pipeline execution completed successfully.")
```